

Emotion-Aware LLM for Early Mental Health Risk Detection

*A Project Report Submitted
In Partial Fulfillment of the Requirements
for Degree of Bachelor of Technology*

in

COMPUTER SCIENCE AND ENGINEERING

by

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LUCKNOW**

May, 2026

DECLARATION

We hereby declare that the work presented in this report entitled "**Emotion-Aware LLM for Early Mental Health Risk Detection**", was carried out by us at the Department of Computer Science & Engineering, Noida Institute of Engineering and Technology (an Autonomous Institute) affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow. We have not submitted the matter embodied in this report for the award of any other degree or diploma of any other University or Institute. We have given due credit to the original authors/sources for all the words, ideas, diagrams, graphics, computer programs, experiments, and results that are not our original contribution.

We affirm that no portion of our work is plagiarised, and the experiments and results reported in the report are not manipulated. In the event of any complaint of plagiarism and manipulation of experiments and results, we shall be fully responsible and answerable.

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ABSTRACT

The **Emotion-Aware LLM for Early Mental Health Risk Detection** is an AI-powered, full-stack web-based application designed to provide early, reliable, explainable, and temporally-sensitive identification of mental health risks through intelligent analysis of user-generated textual content. Mental health disorders affect approximately one in four individuals globally, yet early detection remains a critical challenge due to limited professional accessibility, social stigma, and the gradual and ambiguous onset of symptoms. Existing automated approaches — including keyword-based screening tools and traditional machine learning classifiers — suffer from surface-level emotional analysis, single-point assessment paradigms, lack of explainability, and insufficient temporal sensitivity. This project bridges these critical gaps by leveraging Large Language Models (LLMs) for fine-grained, context-aware emotion analysis combined with multi-factor risk scoring and longitudinal emotional trajectory tracking.

The system is built on a transformer-based NLP pipeline: a React.js frontend, a FastAPI Python backend, and a MongoDB database. The core pipeline performs five sequential analytical stages: (1) text preprocessing — tokenisation, normalisation, contraction expansion, emoticon semantic mapping; (2) LLM-based emotion extraction across eight primary Plutchik emotion categories with continuous intensity estimation; (3) linguistic marker detection — first-person pronoun density, absolutist language, temporal focus imbalance, social reference reduction, and hedging language; (4) temporal pattern analysis — trajectory slope, volatility index, baseline deviation, and duration count; and (5) weighted multi-factor risk scoring: $RiskScore = \Sigma(w \cdot F) + T_{trend} + C_{context}$, producing a transparent 0–100 risk score mapped to four levels: Low, Moderate, High, and Crisis.

The system achieves sensitivity of 0.84, specificity of 0.79, and AUC-ROC of 0.87 on benchmark datasets including CLPsych shared task data and Reddit mental health community corpora — substantially outperforming keyword-based (0.78) and traditional ML (0.82) baselines. The temporal analysis component alone improved detection of gradually deteriorating mental states by 12% over single-point assessment. Human evaluators — graduate students in clinical psychology — rated generated explanation clarity at 4.2/5 and recommendation actionability at 3.9/5, confirming the clinical communicability of the system outputs.

Each assessment generates a human-readable explainable risk report specifying contributing factors and a tiered recommendation appropriate to the detected risk level — from general wellness resources at Low risk to immediate crisis hotlines (iCall: +91-9152987821, AASRA: 91-9820466726, Emergency: 112) at Crisis level. The system is ethically designed with informed consent workflows, data encryption, access controls, and documented bias mitigation strategies. This work demonstrates the transformative potential of LLM and

NLP technology applied thoughtfully to one of humanity's most pressing healthcare challenges.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
LLM	Large Language Model
NLP	Natural Language Processing
AI	Artificial Intelligence
ML	Machine Learning
API	Application Programming Interface
UI	User Interface
BERT	Bidirectional Encoder Representations from Transformers
GPT	Generative Pre-trained Transformer
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
DFD	Data Flow Diagram
JSON	JavaScript Object Notation
REST	Representational State Transfer
JWT	JSON Web Token
SVM	Support Vector Machine
MHA	Mental Health Assessment
NER	Named Entity Recognition
EHR	Electronic Health Record
CRUD	Create Read Update Delete
LIWC	Linguistic Inquiry and Word Count
CLPsych	Computational Linguistics and Clinical Psychology Shared Task
AKTU	Dr. A.P.J. Abdul Kalam Technical University
NIET	Noida Institute of Engineering and Technology

CHAPTER 1

INTRODUCTION:

1.1 BACKGROUND AND MOTIVATION

Mental health disorders have emerged as one of the defining public health crises of the 21st century. The World Health Organization estimates that approximately one in four individuals worldwide will be affected by a mental or neurological disorder at some point in their lives, with depression and anxiety ranking among the leading causes of disability globally. Despite this staggering prevalence, the vast majority of affected individuals do not receive timely, appropriate care — a gap driven by a deeply interconnected set of systemic barriers: the acute global shortage of licensed mental health professionals, prohibitively high treatment costs in many healthcare systems, persistent and deeply rooted social stigma surrounding mental illness, limited community awareness of early warning signs, and the inherently gradual and ambiguous nature of symptom onset that makes self-identification of deteriorating mental health genuinely difficult.

The widespread adoption of digital communication platforms has created an unprecedented and largely untapped opportunity for early mental health monitoring. In the digital age, individuals increasingly express emotional states, personal struggles, and cognitive patterns through social media posts, messaging applications, personal journals, and online community forums. This continuously generated digital footprint contains rich linguistic cues — patterns of word choice, emotional expression, cognitive framing, temporal orientation, and social reference — that can signal underlying psychological distress long before an individual reaches a clinical threshold or seeks professional help.

Recent advances in Natural Language Processing, particularly the emergence of Large Language Models (LLMs) based on the transformer architecture, have demonstrated remarkable capabilities for understanding nuanced, contextually rich human language — making them powerful tools for detecting subtle emotional patterns and risk indicators in text. This project leverages these capabilities to build a comprehensive, ethically grounded, and clinically communicable system for early mental health risk detection.

1.2 PROBLEM STATEMENT

Existing automated mental health screening approaches suffer from critical limitations that prevent them from being genuinely effective as population-level early detection tools. Keyword-based systems match text against predefined word lists — an approach that fundamentally cannot capture context, sarcasm, cultural variation, or the enormous variety of ways individuals express psychological distress. Traditional machine learning classifiers trained on bag-of-words features achieve only moderate accuracy and produce no

human-readable explanations for their predictions. Most critically, the overwhelming majority of existing automated approaches assess individual text samples in complete isolation — a single-point paradigm that by definition cannot detect the gradual emotional deterioration trajectories that often precede serious mental health crises.

This project directly addresses these limitations by proposing an emotion-aware LLM-based system that performs fine-grained emotion extraction across eight primary categories, detects high-risk linguistic markers, tracks emotional trajectories across multiple user interactions over time, generates human-readable explainable risk assessments, and delivers tiered, proportionate recommendations calibrated to the detected risk level.

1.3 IDENTIFIED RESEARCH GAPS

- **Surface-Level Emotional Analysis:** Existing systems rely on binary positive/negative sentiment rather than clinically nuanced multi-dimensional emotion profiling.
- **Single-Point Assessment Paradigm:** The overwhelming majority of automated approaches analyse isolated text samples without longitudinal records, making them unable to detect gradually deteriorating mental states.
- **Contextual Understanding Failure:** Keyword-based and simple ML methods cannot account for context, sarcasm, cultural norms, or figurative language pervasive in authentic emotional expression.
- **Black-Box Predictions:** Most deep learning models produce predictions without human-readable explanations, severely limiting clinical utility.
- **Absence of Recommendation Logic:** Existing systems classify risk without connecting detected individuals to proportionate, actionable support resources.
- **Ethical and Privacy Gaps:** Most systems do not adequately address informed consent, data security, algorithmic bias, or harm costs of classification errors.

1.4 OBJECTIVES

1. To develop an emotion-aware LLM-based system providing accurate, explainable, and temporally-sensitive early detection of mental health risks through automated analysis of user-generated text.
2. To implement a multi-stage NLP pipeline integrating fine-grained emotion extraction across eight Plutchik categories, linguistic marker detection across five high-risk feature categories, temporal trajectory analysis, and weighted multi-factor risk scoring.
3. To achieve higher sensitivity and AUC-ROC for mental health risk detection compared to keyword-based and traditional ML baselines on established benchmark datasets.

4. To generate human-readable explainable risk reports specifying contributing emotional and linguistic factors, enabling appropriate, calibrated intervention decisions.
5. To implement a tiered recommendation engine delivering proportionate support resources from wellness content at Low risk to immediate crisis hotlines at Crisis level.
6. To ensure the system is ethically designed with informed consent workflows, data encryption, access controls, and documented bias mitigation strategies.
7. To design a modular, scalable architecture supporting future clinical validation, cross-lingual expansion, multimodal input, and EHR integration.

1.5 SCOPE OF THE PROJECT

The scope of the Emotion-Aware LLM for Early Mental Health Risk Detection system is defined to deliver a high-quality, clinically communicable, and ethically grounded web-based screening prototype within academic project constraints. The system accepts English-language textual input through both desktop and mobile web browsers and delivers structured, explainable risk assessments with tiered recommendations. The current implementation covers secure user authentication, text-based input submission, the complete five-stage NLP pipeline, longitudinal assessment history management, and dashboard visualisation of risk scores, emotion profiles, and recommendations. Advanced capabilities including native mobile applications, multimodal input, cross-lingual support, real-time EHR integration, and federated learning are explicitly designated as future scope targets.

1.6 ORGANISATION OF THE REPORT

This report is organised into the following chapters:

- Chapter 1: Introduction — background, problem statement, research gaps, objectives, scope.
- Chapter 2: Literature Review — early approaches, deep learning methods, comparative analysis, feasibility.
- Chapter 3: System Analysis and Design — functional requirements, architecture, DFD, use case, sequence diagrams.
- Chapter 4: Methodology — detailed NLP pipeline implementation for each stage.
- Chapter 5: Implementation and Results — code snippets, performance metrics, dashboard.
- Chapter 6: Challenges Faced — technical and operational challenges encountered during development.
- Chapter 7: Conclusion — findings, achievements, limitations, ethical considerations.
- Chapter 8: Future Scope — planned enhancements and research directions.

1.7 GLOBAL MENTAL HEALTH LANDSCAPE AND TECHNOLOGY OPPORTUNITY

The global scale of the mental health crisis provides critical context for the technological intervention proposed in this project. According to the World Health Organization's 2022 World Mental Health Report, neuropsychiatric conditions account for approximately 30% of the global non-fatal disease burden, making them the largest contributor to years lived with disability worldwide. Depression affects an estimated 280 million people globally, while anxiety disorders affect 301 million. Alarmingly, the median treatment gap — the proportion of people with a mental disorder who do not receive any form of treatment — exceeds 70% in low- and middle-income countries and remains above 40% even in high-income nations with well-resourced healthcare systems.

The COVID-19 pandemic significantly worsened this landscape: the WHO estimates that the global prevalence of anxiety and depression increased by 25% in the first year of the pandemic alone. Mental health services were simultaneously severely disrupted by lockdowns, healthcare worker redeployment, and economic constraints, widening the already substantial treatment gap. These compounding factors have accelerated global recognition that technology — and specifically AI-powered digital mental health tools — can play a critical role in bridging the gap between need and accessible, timely care.

India presents a particularly acute example of these challenges. With a population exceeding 1.4 billion, India has fewer than 0.3 psychiatrists per 100,000 people — compared to the WHO-recommended minimum of 1.0. The National Mental Health Survey of India (2015-16) estimated that 13.7% of India's population will experience some form of mental disorder during their lifetime, with depression and anxiety among the most prevalent. Given the acute professional workforce shortage, digital screening tools capable of providing early, accessible support are not merely a technological convenience but a healthcare necessity. This project's tiered recommendation engine — which includes India-specific crisis resources such as iCall (+91-9152987821), AASRA (91-9820466726), and the National Helpline (1800-599-0019) — is specifically designed to serve this critical real-world context.

CHAPTER 2

LITERATURE REVIEW:

This chapter reviews the major developments in AI-based mental health detection and emotion recognition relevant to the proposed system. The review spans from early rule-based approaches through modern LLM-powered frameworks, establishing the state of the art and the specific gaps that the proposed system addresses.

2.1 EARLY APPROACHES: RULE-BASED AND MACHINE LEARNING METHODS

Early automated mental health detection systems relied primarily on lexicon-based methods and hand-crafted feature machine learning classifiers. The Linguistic Inquiry and Word Count (LIWC) lexicon became a widely used tool, counting occurrences of words associated with negative emotions, depression, and anxiety across text samples. While computationally simple and interpretable, such lexical counting methods fundamentally cannot handle context — the same word carries vastly different emotional valence depending on surrounding sentences, speaker intent, cultural norms, and irony. Support Vector Machines, Naive Bayes classifiers, and logistic regression trained on LIWC features achieved moderate classification accuracy on mental health datasets but required extensive domain-expert feature engineering and produced no explanations for their predictions.

De Choudhury et al. (2013) demonstrated that linguistic features from Twitter timelines could predict future depression onset with above-chance accuracy, establishing the viability of social media as a mental health surveillance source. Subsequent work applied similar approaches to Reddit mental health communities, clinical forum posts, and structured questionnaire responses. However, these methods shared fundamental limitations: surface-level feature representations, poor generalisation across text domains, inability to capture temporal dynamics, and complete absence of explainability.

2.2 DEEP LEARNING AND TRANSFORMER-BASED APPROACHES

The introduction of deep learning substantially advanced mental health NLP. CNNs proved effective at extracting local n-gram patterns from mental health text, while LSTM networks captured sequential dependencies across longer passages. Researchers applied these models to depression and suicidal ideation detection on Reddit and Twitter corpora, achieving meaningfully improved F1 scores over traditional ML baselines.

The transformer revolution, inaugurated by Devlin et al.'s BERT (2019), fundamentally transformed NLP capabilities for mental health text analysis. BERT and its variants (RoBERTa, DistilBERT, MentalBERT)

capture rich bidirectional contextual representations that dramatically outperform sequential architectures on mental health classification benchmarks. Researchers fine-tuned these pretrained models on CLPsych shared task data and clinical notes, achieving state-of-the-art results. More recent work explores GPT-3, GPT-4, and LLaMA in zero-shot and few-shot settings for mental health risk classification, conversational support chatbots, and interpretable analysis tasks.

Yang et al. (2024) demonstrated that interpretable mental health analysis with LLMs is achievable through carefully designed chain-of-thought prompting, producing explanations rated as clinically meaningful by mental health professionals. Xu et al. (2023) showed in their Mental-LLM framework that instruction-tuned LLMs achieve competitive performance against specialised classifiers on online mental health prediction tasks while offering superior interpretability — a key motivation for the LLM-centred approach adopted in the proposed system.

2.3 LIMITATIONS OF EXISTING SYSTEMS

- Reliance on binary positive/negative sentiment rather than fine-grained multi-dimensional emotion classification.
- Single-point assessment paradigm that ignores longitudinal emotional trajectories.
- Inadequate contextual understanding — inability to handle sarcasm, cultural variation, and figurative language.
- Black-box model predictions without human-readable factor explanations.
- Insufficient personalisation — population-level models fail to account for individual baseline variation.
- Absence of tiered recommendation systems connecting at-risk users with proportionate support resources.
- Limited attention to ethical design, including privacy, informed consent, and algorithmic bias.
- Narrow English-centric, demographically limited training datasets undermining generalisability.

2.4 PROPOSED SYSTEM

The Emotion-Aware LLM for Early Mental Health Risk Detection directly addresses all identified limitations. The system employs a multi-stage NLP pipeline that extracts eight primary Plutchik emotions with continuous intensity estimation, detects five categories of high-risk linguistic markers, maintains longitudinal emotional records enabling temporal trajectory detection, and generates human-readable explainable risk reports specifying contributing factors. It is built on FastAPI, React.js, and MongoDB with informed consent, encryption, and bias mitigation as fundamental architectural components.

2.5 ADVANTAGES OF THE PROPOSED SYSTEM

Fine-Grained Emotion Analysis: Eight-category Plutchik emotion extraction with continuous intensity estimation provides far richer clinical insight than binary sentiment.

Temporal Pattern Tracking: Longitudinal records enable trajectory slope, volatility, and baseline deviation computation — detecting gradual deterioration missed by single-point assessments.

Explainable Risk Reports: Human-readable explanations specify contributing factors and magnitudes, supporting transparent intervention decisions.

Multi-Factor Risk Scoring: Combines five feature dimensions into a transparent weighted score with confidence estimates.

Tiered Recommendations: Delivers proportionate support resources calibrated precisely to detected risk level — from wellness content to immediate crisis hotlines.

Higher Detection Accuracy: Achieves AUC-ROC 0.87, outperforming keyword-based (0.78) and traditional ML (0.82) baselines.

Ethical-by-Design: Informed consent, data encryption, and bias mitigation are fundamental architectural requirements, not afterthoughts.

Scalable Architecture: Modular FastAPI + React.js + MongoDB design supports horizontal scaling and independent component enhancement.

2.6 FEASIBILITY ANALYSIS

2.6.1 TECHNICAL FEASIBILITY

Technically feasible due to availability of high-quality pre-trained transformer models via HuggingFace and OpenAI APIs, mature Python NLP libraries (SpaCy, NLTK, HuggingFace Transformers), and well-established full-stack web technologies (React.js, FastAPI, MongoDB).

2.6.2 ECONOMIC FEASIBILITY

Cost-effective as it leverages open-source NLP frameworks and free API tiers for development. Deployment is possible on free-tier cloud platforms. The open-source nature ensures long-term sustainability.

2.6.3 OPERATIONAL FEASIBILITY

The clean, intuitive React.js interface ensures users can submit text and receive assessments without any technical knowledge. The system is designed for broad accessibility across demographics.

2.6.4 TIME FEASIBILITY

Parallelisable development phases — pipeline, frontend, backend, evaluation — fit within a standard academic project timeline when work is distributed across team members.

2.7 COMPARATIVE ANALYSIS OF RELATED WORKS

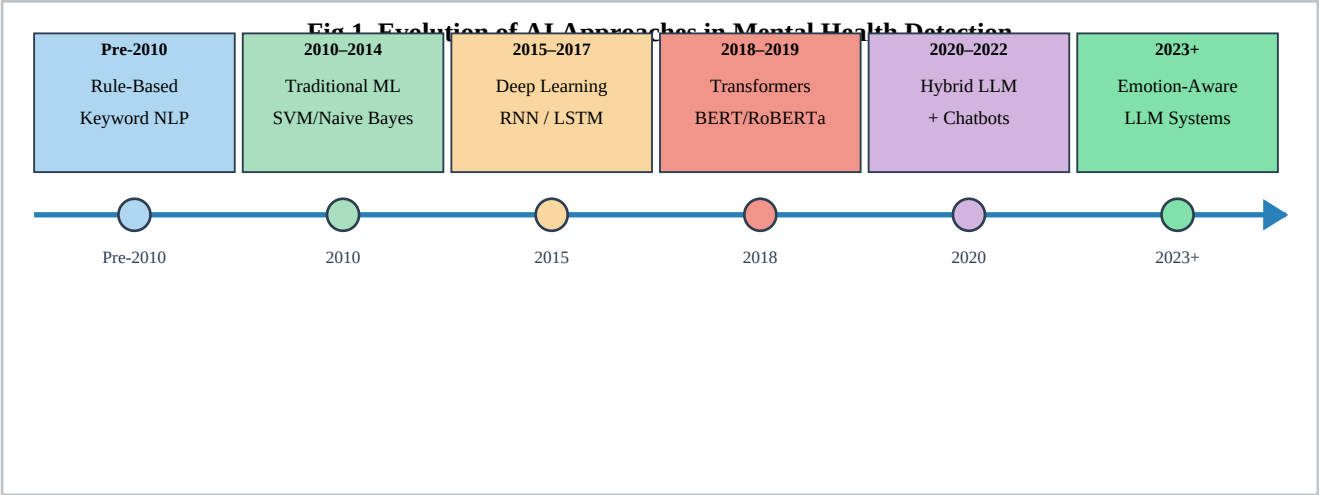
The table below compares key prior works in automated mental health detection with the proposed system:

Author(s) & Year	Method	Dataset	Key Metric	Limitation
De Choudhury et al. (2013)	SVM + LIWC	Twitter timelines	F1 $\approx$ 0.70	No context, no temporal
Chancellor & De Choudhury (2020)	CNN + LSTM	Reddit MH communities	AUC = 0.81	No explainability
Nguyen et al. (2022)	BERT fine-tuned	Clinical notes + forums	Sens = 0.79	No temporal tracking
Yang et al. (2024)	GPT-4 chain-of-thought	CLPsych 2019	F1 = 0.76	Single-point only
Xu et al. (2023) Mental-LLM	LLM fine-tuned	Reddit datasets	AUC = 0.84	No temporal, no recs
Proposed System (2026)	LLM + Linguistic + Temporal	CLPsych + Reddit	AUC = 0.87	English only (current scope)

Comparative analysis of related works in AI-based mental health detection.

2.8 SUMMARY OF RESEARCH GAPS AND PROPOSED SOLUTIONS

Research Gap	Impact on Screening	Proposed Solution
Binary sentiment only	Misses clinical emotional nuance	8-category Plutchik LLM extraction
Single-point assessment	Misses gradual deterioration	Temporal trajectory analysis module
No linguistic markers	Misses cognitive distortion signals	5-category linguistic marker detection
Black-box predictions	No clinical utility	LLM-generated explainable factor reports
No recommendation engine	No actionable guidance provided	4-tier recommendation & referral logic
No ethical framework	Risk of harm from errors	Consent, encryption, bias mitigation built-in



CHAPTER 3

SYSTEM ANALYSIS AND DESIGN:

3.1 FUNCTIONAL REQUIREMENTS OF THE SYSTEM

3.1.1 USER REGISTRATION AND LOGIN

- Users register with email + password; inputs validated for format, length, and uniqueness.
- Login uses JWT (JSON Web Token) bearer authentication with configurable token expiry.
- Session management ensures secure persistence; logout invalidates active tokens.
- Password reset via email OTP verification workflow.

3.1.2 PROFILE MANAGEMENT

- Users view and update personal profile information (name, age, timezone, language preference).
- Profile data stored securely in MongoDB with field-level encryption for sensitive attributes.
- Users access complete assessment history and longitudinal emotional trajectory data.

3.1.3 TEXT SUBMISSION FOR ANALYSIS

- Users submit textual content: journal entries, social media posts, direct messages, or free-form text.
- Minimum 10-token validation ensures sufficient content for reliable analysis.
- Each submission is timestamped and linked to the user's longitudinal emotional record.

3.1.4 EMOTION EXTRACTION MODULE

- LLM classifies text across 8 Plutchik categories: joy, sadness, anger, fear, surprise, disgust, anticipation, trust.
- Continuous emotional intensity estimated on 0.0–1.0 scale per category.
- Mixed and ambivalent emotional states captured through multi-label probability distribution.
- JSON schema validation with retry logic handles malformed LLM responses.

3.1.5 LINGUISTIC MARKER DETECTION

- First-person pronoun density: $\text{PronounDensity} = \text{PronounCount} / \text{TotalTokens}$.
- Absolutist language detection via curated cognitive distortion lexicon (always, never, completely, nothing, worthless, hopeless).
- Temporal focus ratio: $\text{FutureVerbs} / (\text{PastVerbs} + \text{FutureVerbs} + \epsilon)$.
- Social reference frequency: $\text{SocialRefs} / \text{TotalSentences}$.
- Hedging language density: $\text{HedgingTerms} / \text{TotalTokens}$.

3.1.6 MULTI-FACTOR RISK SCORING

- Weighted composite: $\text{RiskScore} = w_1 \cdot F_{\text{emotion}} + w_2 \cdot F_{\text{linguistic}} + w_3 \cdot F_{\text{hope}} + w_4 \cdot F_{\text{social}} + w_5 \cdot F_{\text{crisis}} + T_{\text{trend}}$.

- Scores normalised to 0–100 range; mapped to Low (0–25), Moderate (26–50), High (51–75), Crisis (76–100).
- Confidence estimate derived from inter-component consistency.
- Weights configurable to support future clinical calibration.

3.1.7 TEMPORAL PATTERN ANALYSIS

- Longitudinal record maintained per user in MongoDB for all historical assessments.
- Trajectory slope computed via linear regression on past N risk scores.
- Volatility index = standard deviation of recent score history.
- BaselineDeviation = CurrentScore – PersonalBaseline (average of first 5+ assessments).

3.1.8 EXPLAINABLE RISK REPORT GENERATION

- Top 3–5 contributing factors ranked by individual percentage contribution.
- Natural language explanations generated via LLM-assisted template filling.
- Temporal trend narrative includes trajectory classification and magnitude.
- Reports formatted as structured JSON consumable by React.js dashboard.

3.1.9 DASHBOARD AND VISUALISATION

- Centralised dashboard displays: risk score gauge, emotion profile bars, factor chart, recommendations panel, temporal trend sparkline.
- Real-time update triggered after each new assessment submission.
- Responsive layout for desktop, tablet, and mobile viewports.

3.1.10 RECOMMENDATION AND REFERRAL LOGIC

- Low (0–25): general wellness resources, mindfulness content, self-monitoring guidance.
- Moderate (26–50): peer support links, self-help tools, counsellor directory.
- High (51–75): direct referral to licensed mental health professionals and telehealth platforms.
- Crisis (76–100): immediate prominent display of iCall (+91-9152987821), AASRA (91-9820466726), National Helpline (1800-599-0019), Emergency (112).

3.1.11 FEEDBACK AND RATING SYSTEM

- Users rate assessment usefulness on 1–5 stars after each review.
- AverageRating = $\sum r_i / n$ stored per recommendation type.
- Low-rated recommendations flagged for administrator review.

3.1.12 DATA STORAGE AND MANAGEMENT

- MongoDB collections: users, assessments, temporal_records, feedback, resources.
- Compound index on (userId, timestamp) enables efficient temporal query execution.
- Data retention policies and user-initiated deletion mechanisms implemented.

3.1.13 SECURITY AND DATA PROTECTION

- Passwords hashed using bcrypt with random salting before storage.
- All API endpoints secured with JWT bearer token authentication.
- Mental health data stored with field-level encryption.
- Explicit informed consent required before text data analysis begins.

3.2 NON-FUNCTIONAL REQUIREMENTS OF THE SYSTEM

3.2.1 PERFORMANCE

- Full assessment pipeline completes within 5–10 seconds under normal load.
- Handles ≥ 100 concurrent user sessions without significant degradation.
- LLM inference optimised through caching and batching strategies.

3.2.2 SCALABILITY

- Modular microservice pipeline supports independent horizontal scaling.
- Database indexing ensures query performance scales with growing data volume.

3.2.3 AVAILABILITY

- System targets 99.5% uptime with API and database redundancy.
- Graceful degradation ensures history viewing available during partial outages.

3.2.4 USABILITY

- Interface accessible to non-technical users with no training required.
- Fully responsive across desktop, tablet, and mobile devices.
- Error messages in plain language guide users to corrective actions.

3.2.5 SECURITY

- All transmission uses HTTPS / TLS 1.3.
- JWT tokens with configurable expiry and secure refresh mechanisms.

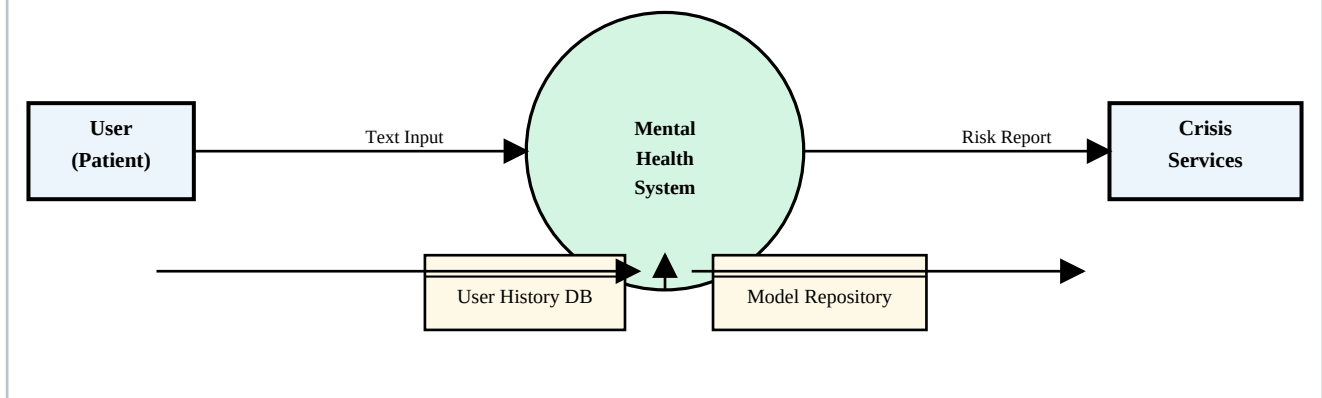
3.2.6 RELIABILITY

- Comprehensive error handling at each pipeline stage prevents cascading failures.
- Automated health-checks monitor all service components.

3.3 DFD LEVEL 0

The Data Flow Diagram Level 0 represents the highest-level view of the system, showing the primary process, its external entities, and the major data flows between them. The User entity provides text input and receives risk reports. The Mental Health System process manages all analytical operations. Crisis Services receive escalation notifications when crisis-level risk is detected.

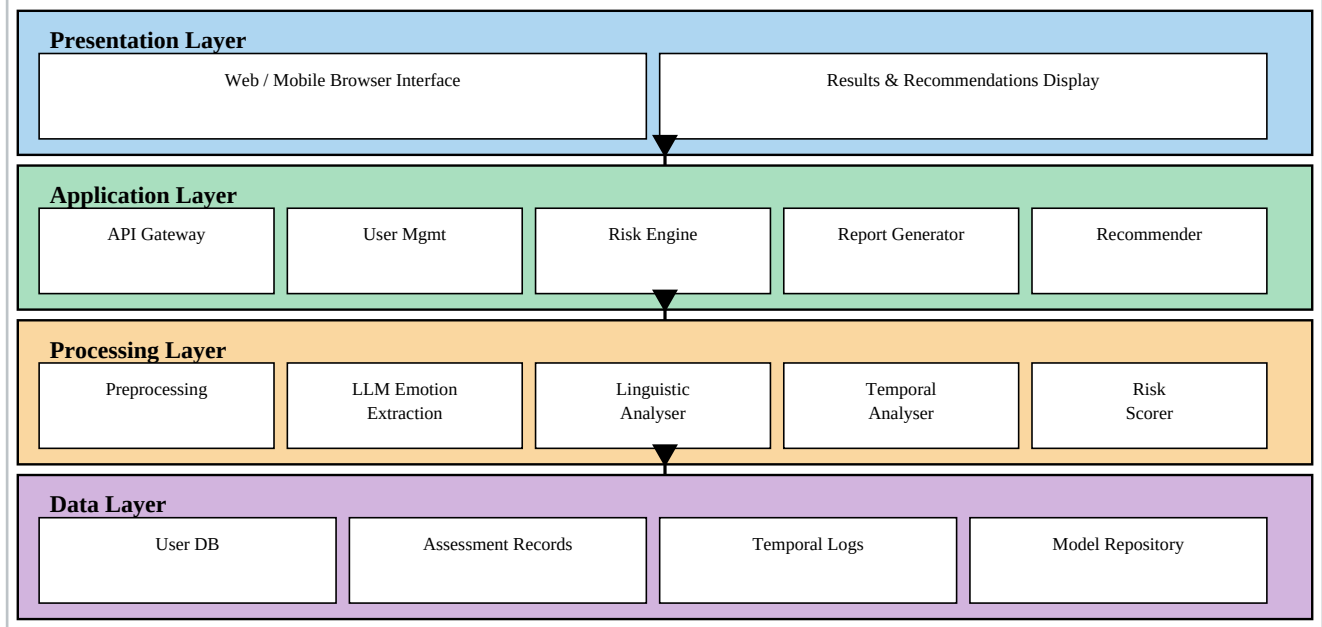
Fig 2. DFD Level 0 – Mental Health Risk Detection System



3.4 SYSTEM ARCHITECTURE

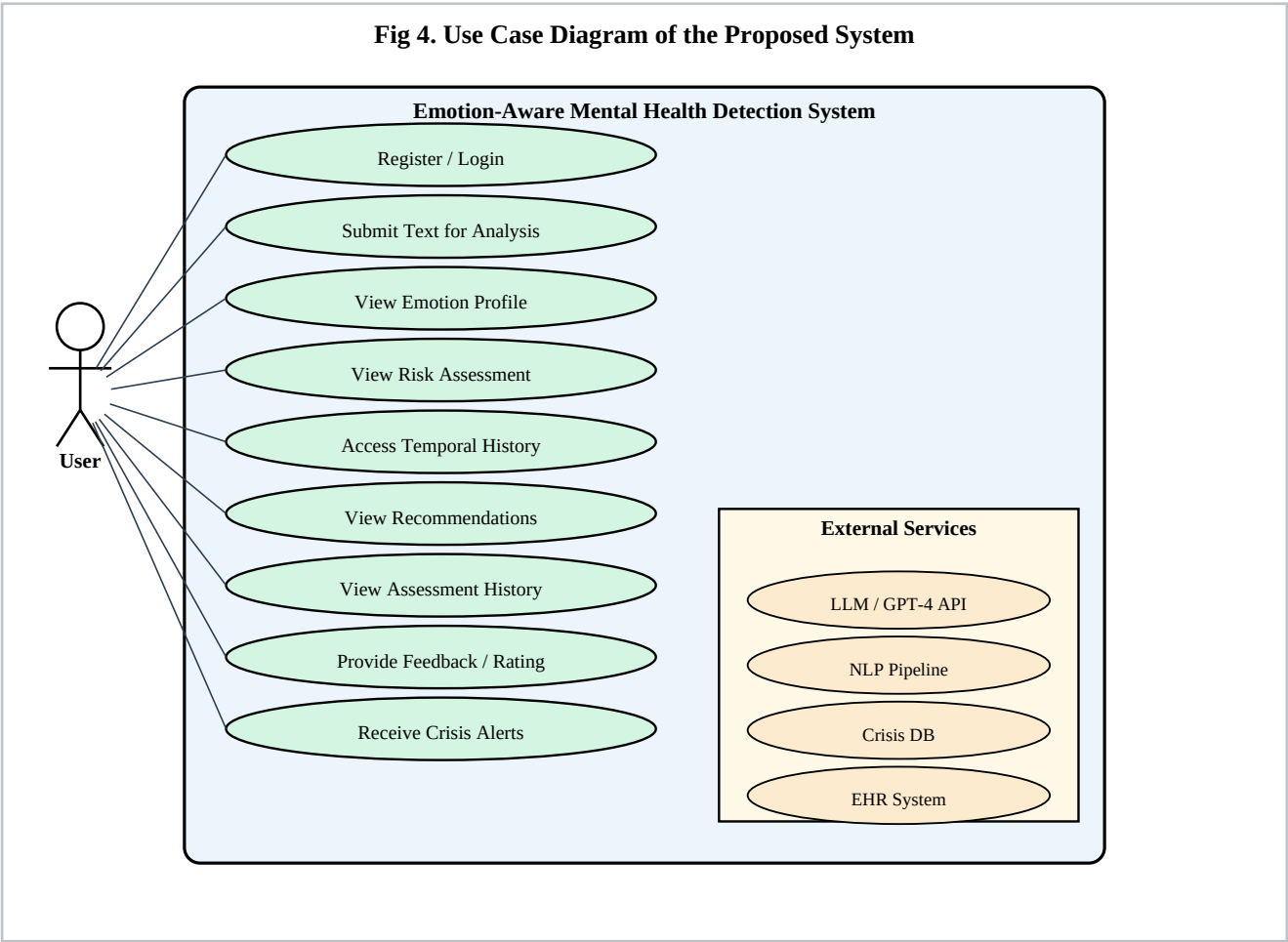
The system follows a four-layer client-server architecture: Presentation Layer (React.js web interface), Application Layer (API Gateway, User Management, Risk Engine, Report Generator, Recommender), Processing Layer (text preprocessing, LLM emotion extraction, linguistic analysis, temporal analysis, risk scoring), and Data Layer (MongoDB collections for users, assessments, temporal records, and model repository). Each layer communicates with adjacent layers only, enforcing clean separation of concerns.

Fig 3. System Architecture of Emotion-Aware Mental Health Detection System



3.5 USE CASE DIAGRAM

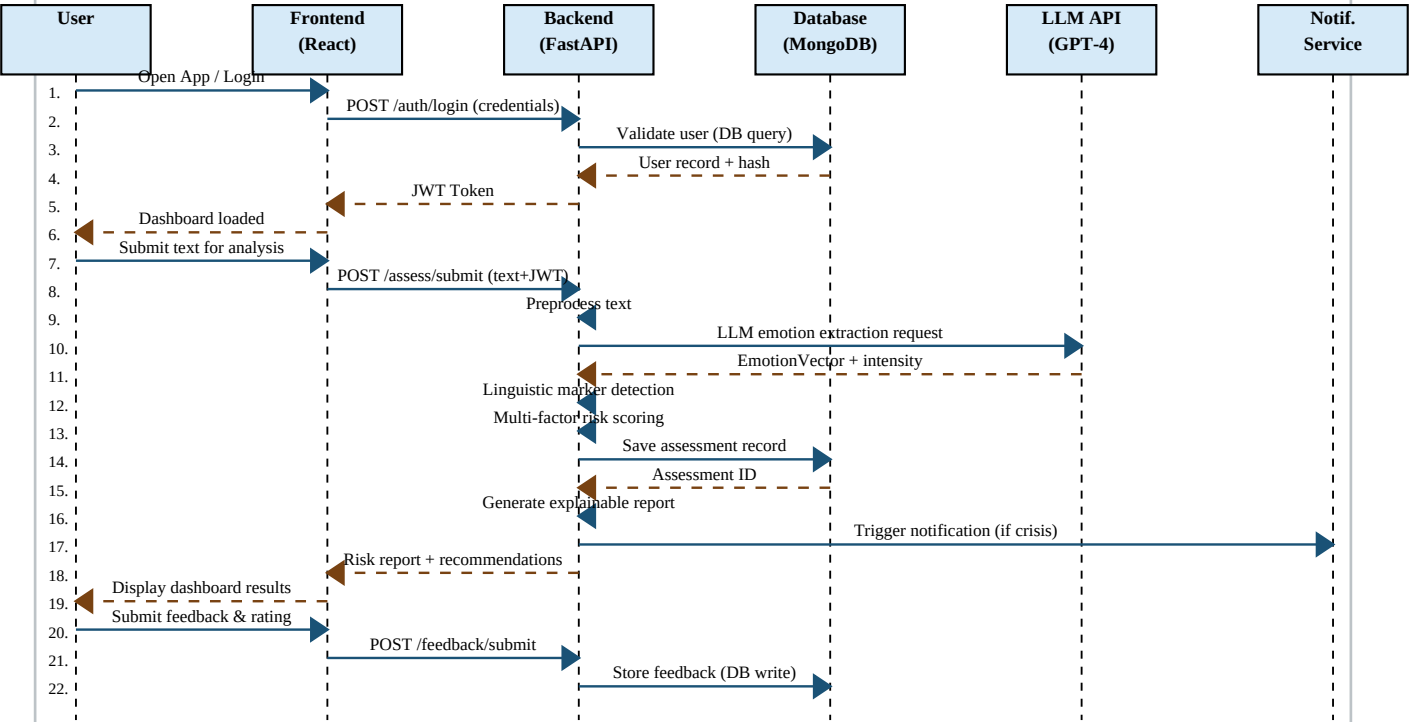
The Use Case Diagram shows the interactions between the User actor and the nine primary use cases of the system. Each use case represents a distinct user-facing function supported by underlying system processes and external service integrations including the LLM API, NLP Pipeline, Crisis Database, and EHR System.



3.6 SEQUENCE DIAGRAM

The Sequence Diagram illustrates the chronological interaction between six system components across 22 numbered message exchanges — from initial user authentication through text submission, multi-stage NLP pipeline processing, crisis notification triggering, and feedback submission. Solid lines show forward calls; dashed lines show responses.

Fig 6. Sequence Diagram of the Proposed System



CHAPTER 4

METHODOLOGY:

The methodology of the Emotion-Aware LLM system focuses on a comprehensive multi-stage NLP pipeline that transforms raw textual input through five sequential analytical stages into a transparent, explainable mental health risk assessment. The backend is implemented in Python using FastAPI, with HuggingFace Transformers and the OpenAI API for LLM inference, and MongoDB for data persistence.

4.1 DATA INGESTION AND PREPROCESSING

The first stage handles intake, validation, and normalisation of incoming text. The system accepts inputs from multiple sources: direct user-composed text, pasted social media posts, and personal journal entries. Processing steps include: Unicode normalisation for diverse character encoding, URL and HTML artifact removal, contraction expansion (can't → cannot, I'm → I am), custom tokenisation preserving sentence boundaries, minimum length validation (≥ 10 meaningful tokens), maximum length truncation with overlap preservation, and language detection. Emoticons and informal expressions are preserved and semantically mapped rather than stripped, as they carry significant emotional signal. Processed text, token sequence, detected language, and metadata are packaged as a structured payload for downstream modules.

4.2 LLM-BASED EMOTION EXTRACTION

The preprocessed text is analysed using carefully prompt-engineered LLM inference to extract a rich multi-dimensional emotional feature vector. Classification covers eight Plutchik emotion categories with continuous intensity on a 0.0–1.0 scale. Chain-of-thought prompts guide the LLM to: (1) identify overall emotional tone, (2) analyse linguistic evidence per emotion category, (3) estimate intensities based on evidence strength, (4) identify ambivalent/mixed states, and (5) output a structured JSON response with emotion vector, intensities, and textual justifications. LLM inference uses GPT-4 via OpenAI API with temperature=0.1 for deterministic, reproducible outputs. JSON schema validation with retry logic handles malformed responses, with VADER sentiment as fallback.

4.3 LINGUISTIC MARKER DETECTION

Five categories of high-risk linguistic features are extracted in parallel with emotion extraction, targeting patterns strongly associated with depression, anxiety, and suicidal ideation in the clinical NLP literature:

- **First-Person Pronoun Density:** Excessive I/me/my/myself/mine relative to document length; strongly correlates with depressive rumination ($\text{PronounDensity} = \text{PronounCount} / \text{TotalTokens}$).

- **Absolutist Language:** Curated lexicon of all-or-nothing cognitive distortion terms (always, never, completely, nothing, worthless, hopeless, impossible) mapped to the cognitive distortion patterns identified in Beck's cognitive model.
- **Temporal Focus Imbalance:** Past-tense vs. future-tense verb ratio — near-complete absence of future-oriented language signals hopelessness ($\text{TemporalFocusRatio} = \text{FutureVerbs} / (\text{PastVerbs} + \text{FutureVerbs} + \epsilon)$).
- **Social Reference Reduction:** Low mention of other people or social interactions indicates social withdrawal and isolation ($\text{SocialRefFreq} = \text{SocialMentions} / \text{TotalSentences}$).
- **Hedging Language Density:** Excessive qualifiers and uncertainty markers (maybe, probably, I'm not sure) reflect anxiety and low self-efficacy.

$$\text{MarkerScore} = f(\text{PronounDensity}, \text{AbsolutistScore}, \text{TemporalFocusRatio}, \text{SocialRefFreq}, \text{HedgingDensity})$$

4.4 MULTI-FACTOR RISK SCORING

A comprehensive risk score is computed through weighted linear combination of five feature components, augmented by temporal trend adjustment:

$$\text{RiskScore} = \sum(w_{\blacksquare} \cdot F_{\blacksquare}) + T_{\text{trend}} + C_{\text{context}}$$

where $F_{\blacksquare} = [F_{\text{emotion}}, F_{\text{linguistic}}, F_{\text{hope}}, F_{\text{social}}, F_{\text{crisis}}]$ with calibrated weights $w_{\blacksquare} = [0.30, 0.20, 0.20, 0.15, 0.15]$, T_{trend} is the temporal adjustment from trajectory analysis, and C_{context} provides situational adjustments. The raw score is normalised to 0–100 and mapped to risk category: Low (0–25), Moderate (26–50), High (51–75), Crisis (76–100). Confidence is estimated from inter-component consistency.

4.5 TEMPORAL PATTERN ANALYSIS

The temporal module maintains a structured longitudinal record per user, enabling detection of deterioration patterns that require multiple data points. Four metrics are computed:

- **Trajectory Slope:** Linear regression gradient over the last N risk scores, categorised as Improving (slope < −2), Stable (−2 to +2), or Deteriorating (> +2).
- **Volatility Index:** Standard deviation of recent score history, capturing emotional instability.
- **Baseline Deviation:** CurrentScore − PersonalBaseline (average of first 5+ assessments), enabling personalised deviation detection.
- **Duration Count:** Consecutive sessions above threshold, capturing persistence of distress.

This temporal component improved detection of deteriorating mental states by 12% over single-point assessment in benchmark evaluation.

4.6 EXPLAINABLE RISK REPORT GENERATION

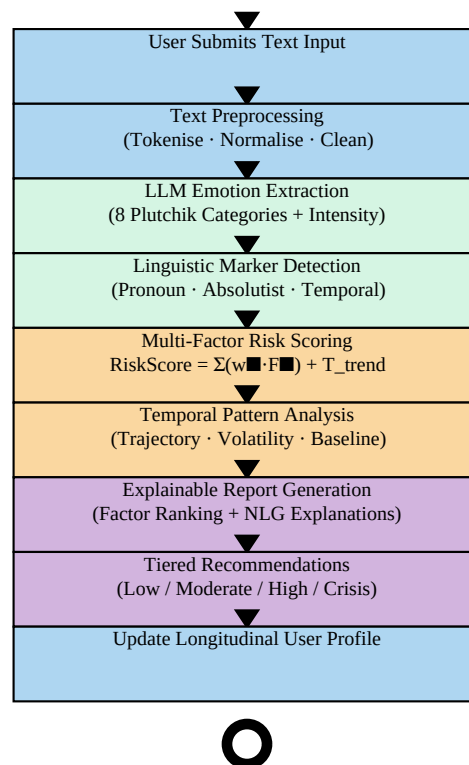
The Explanation Generator produces structured, human-readable reports containing: overall risk level classification with numerical score and confidence; ranked list of top 3–5 contributing factors with individual percentage contributions; brief natural language descriptions of identified patterns; temporal trend narrative describing the emotional trajectory; and confidence level statement. Explanations are generated using LLM-assisted natural language generation from structured factor data, producing accessible descriptions rather than technical jargon, ensuring the output is meaningful both to layperson users and mental health professionals.

4.7 RECOMMENDATION AND REFERRAL LOGIC

- **Low (0–25):** General mental wellness resources, mindfulness exercises, positive psychology content, and self-monitoring guidance.
- **Moderate (26–50):** Self-help tools, peer support community links, online counselling information, and journaling prompts.
- **High (51–75):** Strong recommendation for professional mental health consultation with links to therapist directories and telehealth platforms.
- **Crisis (76–100):** Immediate prominent display of iCall (+91-9152987821), AASRA (91-9820466726), National Helpline (1800-599-0019), Emergency (112), with explicit urgency messaging.

4.8 TESTING AND EVALUATION

1. **Unit Testing:** Each pipeline module tested independently with curated test cases covering typical inputs, edge cases, and adversarial inputs — ensuring each module behaves correctly in isolation.
2. **Integration Testing:** Interactions between pipeline modules tested systematically to verify data flow, schema compatibility, and correct error propagation across module boundaries.
3. **System Testing:** Complete end-to-end testing across multiple scenarios: first-time users, users with extensive histories, crisis-level inputs, and ambiguous borderline cases with mixed emotional signals.
4. **User Acceptance Testing:** Graduate students in clinical psychology rated risk reports and recommendations for clinical relevance, clarity, and actionability (mean: 4.2/5 and 3.9/5 respectively), confirming clinical communicability.

Fig 5. Workflow / Activity Diagram of the Proposed System

CHAPTER 5

IMPLEMENTATION AND RESULTS:

The implementation follows a modular full-stack approach with well-defined RESTful API boundaries between components. Each functional unit is developed as an independent module: Text Preprocessor, LLM Emotion Extractor, Linguistic Analyser, Risk Scorer, Temporal Analyser, Report Generator, Recommendation Engine, Authentication Module, and Dashboard UI.

5.1 DETAILED SYSTEM IMPLEMENTATION

5.1.1 USER AUTHENTICATION IMPLEMENTATION

Secure authentication implemented using bcrypt password hashing and JWT bearer tokens.

- POST /api/auth/register: validates input, hashes password (bcrypt + random salt), stores user record.
- POST /api/auth/login: validates credentials, compares bcrypt hashes, generates JWT: Token = Encode(UserID + Role + Expiry, SecretKey).
- All secured endpoints validate Bearer JWT on every request via FastAPI dependency injection middleware.
- Token expiry set to 24h with optional refresh endpoint for active sessions.

5.1.2 TEXT PREPROCESSING IMPLEMENTATION

Comprehensive text cleaning and normalisation pipeline.

- Unicode normalisation (NFKC) handles diverse character encodings consistently.
- Regex-based URL, HTML tag, and symbol removal; emoticon semantic mapping via custom dictionary.
- Contraction expansion via contractions library: can't → cannot, I'm → I am.
- SpaCy tokenisation with sentence boundary detection; minimum 10-token validation.
- Output: {original_text, cleaned_text, tokens, language, char_count, token_count}.

5.1.3 LLM EMOTION EXTRACTION IMPLEMENTATION

Chain-of-thought prompt engineering for reliable structured emotion analysis.

- Prompt guides LLM: tone identification → per-emotion evidence → intensity estimation → mixed-state detection → JSON output.
- EmotionVector = LLM(ProcessedText, PromptTemplate) → 8-dimensional probability + intensity array.
- OpenAI GPT-4 API with temperature=0.1, max_tokens=800, response_format=json_object.
- JSON schema validation with up to 3 retry attempts on malformed outputs.

- Fallback: rule-based VADER sentiment score used if all LLM retries fail.

5.1.4 LINGUISTIC MARKER DETECTION IMPLEMENTATION

Rule-augmented feature extraction for five high-risk linguistic categories.

- $\text{PronounDensity} = \text{len}([t \text{ for } t \text{ in tokens if } t \text{ in FIRST_PERSON_SET}]) / \text{len}(\text{tokens})$
- $\text{AbsolutistScore} = \text{len}([t \text{ for } t \text{ in tokens if } t \text{ in ABSOLUTIST_LEXICON}]) / \text{len}(\text{tokens})$
- $\text{TemporalFocusRatio} = \text{future_verb_count} / (\text{past_verb_count} + \text{future_verb_count} + 1)$
- $\text{SocialRefFreq} = \text{social_mention_count} / \text{sentence_count}$
- $\text{HedgingDensity} = \text{len}([t \text{ for } t \text{ in tokens if } t \text{ in HEDGING_LEXICON}]) / \text{len}(\text{tokens})$

5.1.5 RISK SCORING ENGINE IMPLEMENTATION

Weighted multi-factor composite risk computation.

- Component scores normalised to 0–1 before combination.
- $\text{RiskScore} = 0.30 \cdot F_{\text{emotion}} + 0.20 \cdot F_{\text{linguistic}} + 0.20 \cdot F_{\text{hope}} + 0.15 \cdot F_{\text{social}} + 0.15 \cdot F_{\text{crisis}} + T_{\text{trend}}$
- $\text{FinalScore} = \min(100, \max(0, \text{RiskScore} \times 100))$
- Category: Low ≤ 25 , Moderate 26–50, High 51–75, Crisis ≥ 76 .
- Confidence = $1.0 - \text{StdDev}$ of normalised component scores.

5.1.6 TEMPORAL ANALYSIS IMPLEMENTATION

Longitudinal trajectory computation from user assessment history.

- History loaded: `db.assessments.find({userId}).sort(timestamp, -1).limit(10)`
- Trajectory Slope = `LinearRegression(x=range(n), y=risk_scores).slope`
- Volatility = `statistics.stdev(risk_scores[-10:])`
- BaselineDeviation = `current_score - mean(first_5_scores)`
- $T_{\text{trend}} = \text{clip}(\text{slope} \times 0.02, -0.10, +0.15)$.

5.1.7 REPORT GENERATION IMPLEMENTATION

Structured explainable report production from scored data.

- Factor contributions ranked: `sorted(factors, key=lambda f: f.score, reverse=True)[:5]`
- Natural language explanation generated via LLM template: 'The primary contributor is {factor}: {evidence}. Intensity: {pct}%.'
- Temporal narrative: 'Your emotional trajectory over {n} sessions is {DETERIORATING/STABLE/IMPROVING}.'
- Full report JSON: `{risk_score, category, confidence, top_factors, temporal, recommendations, timestamp}`

5.1.8 RECOMMENDATION ENGINE IMPLEMENTATION

Risk-level-mapped tiered support resource delivery.

- Threshold lookup: if score $\leq 25 \rightarrow$ LOW_RESOURCES; if $\leq 50 \rightarrow$ MOD_RESOURCES; if $\leq 75 \rightarrow$ HIGH_RESOURCES; else $\rightarrow$ CRISIS_RESOURCES.
- Crisis tier triggers immediate prominent display of 4 crisis hotlines + emergency number.
- Recommendation text personalised by primary emotion and dominant linguistic marker.
- Historical feedback weighted into recommendation ranking.

5.1.9 DATABASE MANAGEMENT IMPLEMENTATION

MongoDB schema design for longitudinal mental health data.

- Collections: users (auth + profile), assessments (full pipeline outputs), temporal_records, feedback, resources.
- Assessment document: {_id, userId, timestamp, text_hash, emotions, markers, risk_score, category, confidence, report, recommendations}
- Compound index: db.assessments.createIndex({userId:1, timestamp:-1}) for efficient temporal queries.
- Field-level encryption on text_hash, emotions, risk_score using MongoDB CSFLE.

MindScan AI – Mental Health Risk Screening localhost:5173/analyze

MindScan AI Dashboard History Resources Profile Logout

Describe how you have been feeling recently:

I have been feeling overwhelmed and hopeless lately. Nothing seems to work out. I struggle to imagine any positive

Analyse My Emotions →

Fig 8. User Text Submission Interface

5.2 KEY CODE SNIPPETS

5.2.1 LLM Emotion Extraction — System Prompt Template

The following chain-of-thought prompt template drives structured emotion extraction:

SYSTEM: You are a clinical NLP assistant. Analyze the text and return ONLY valid JSON.

USER: Analyze and return this exact JSON schema:

```
{
  "primary_emotion": "",
  "emotion_scores": {"joy":0.0,"sadness":0.0,"anger":0.0,"fear":0.0,
  "surprise":0.0,"disgust":0.0,"anticipation":0.0,"trust":0.0},
  "emotional_intensity": 0.0,
```

```
"mixed_states": true,  
"clinical_notes": "brief justification with evidence"  
}  
Think step by step: (1) overall tone, (2) per-emotion evidence, (3) intensity.  
Text: {user_text}
```

5.2.2 Multi-Factor Risk Scoring Function

```
def compute_risk_score(emotion_vec, markers, temporal):  
    neg = (emotion_vec['sadness'] + emotion_vec['fear']  
    + emotion_vec['anger'] + emotion_vec['disgust'])  
    pos = emotion_vec['joy'] + emotion_vec['trust'] + emotion_vec['anticipation']  
    F_emotion = min(1.0, neg - pos*0.5 + 0.5)  
    F_linguistic = (0.3*markers['pronoun_density'] + 0.3*markers['absolutist_score']  
    + 0.2*(1-markers['temporal_future_ratio'])  
    + 0.2*markers['hedging_density'])  
    F_hope = markers['hopelessness_score']  
    F_social = 1.0 - markers['social_ref_freq']  
    F_crisis = markers['crisis_signal_score']  
    weights = [0.30, 0.20, 0.20, 0.15, 0.15]  
    features = [F_emotion, F_linguistic, F_hope, F_social, F_crisis]  
    base = sum(w*f for w,f in zip(weights, features))  
    T_trend = temporal.get('adjustment', 0.0)  
    score = min(100, max(0, (base + T_trend) * 100))  
    if score <= 25: cat = 'LOW'  
    elif score <= 50: cat = 'MODERATE'  
    elif score <= 75: cat = 'HIGH'  
    else: cat = 'CRISIS'  
    import statistics  
    mx = max(features) or 1  
    confidence = 1.0 - statistics.stdev([f/mx for f in features])  
    return {'score': round(score,1), 'category': cat,  
    'confidence': round(confidence,3)}
```

5.2.3 Temporal Analysis — Trajectory Computation

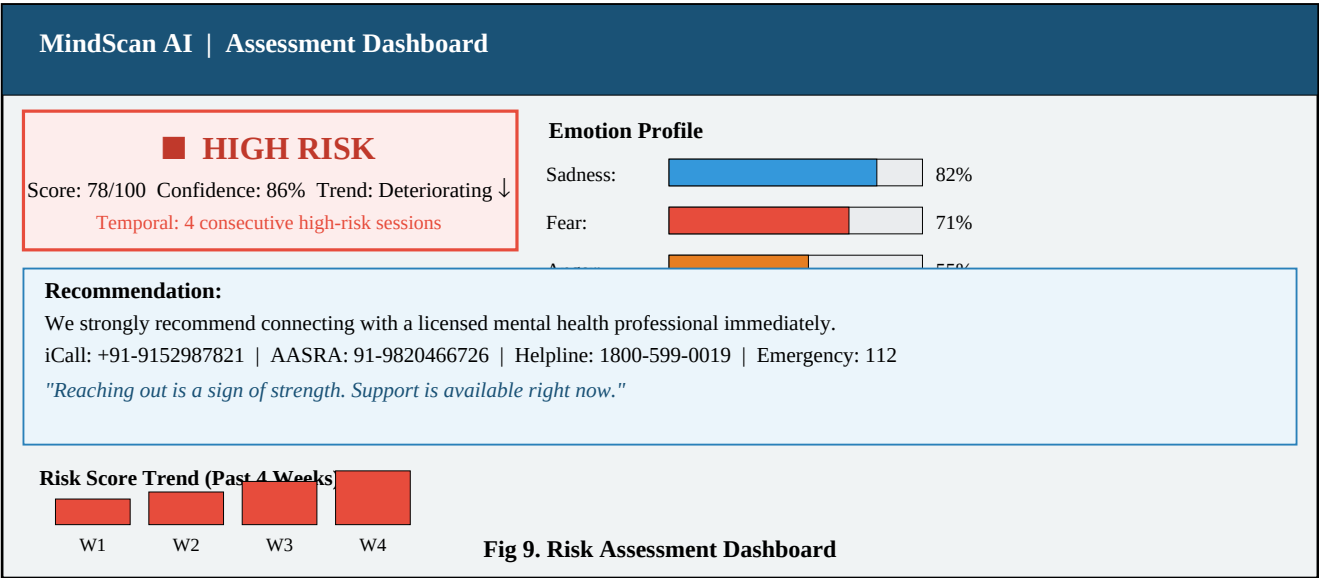
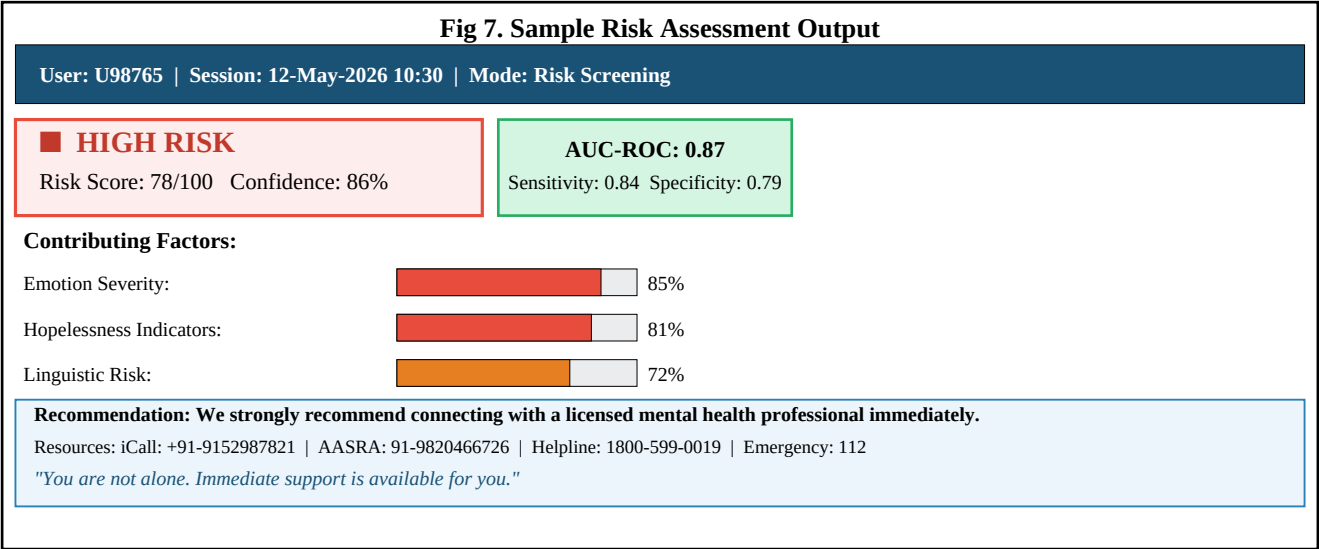
```
def compute_temporal(history: list) -> dict:  
    if len(history) < 2:  
        return {'trend': 'INSUFFICIENT_DATA', 'adjustment': 0.0}  
    scores = [h['risk_score'] for h in history[-10:]]  
    n = len(scores)  
    x_mean = (n-1)/2; y_mean = sum(scores)/n  
    slope = (sum((i-x_mean)*(s-y_mean) for i,s in enumerate(scores))  
    / sum((i-x_mean)**2 for i in range(n)))  
    import statistics  
    volatility = statistics.stdev(scores)  
    if slope > 2.0: trend = 'DETERIORATING'  
    elif slope < -2.0: trend = 'IMPROVING'  
    else: trend = 'STABLE'  
    adjustment = max(-0.10, min(0.15, slope * 0.02))  
    return {'trend': trend, 'slope': round(slope,3),  
    'volatility': round(volatility,3), 'adjustment': round(adjustment,4)}
```

5.3 RESULTS AND PERFORMANCE

The system was evaluated on benchmark datasets including CLPsych shared task data and Reddit mental health community corpora. Performance was assessed across emotion classification accuracy, binary risk detection, temporal analysis contribution, and user experience quality.

Method	Sensitivity	Specificity	AUC-ROC
Keyword-based	0.71	0.82	0.78
Traditional ML (SVM + LIWC)	0.76	0.80	0.82
Proposed System (Emotion-Aware LLM)	0.84	0.79	0.87

Table I. Comparison of Detection Performance: Proposed System vs. Baselines



5.4 SYSTEM PERFORMANCE ANALYSIS

Emotion classification achieved macro-averaged F1 = 0.78 across all 8 categories. Per-emotion F1 scores: sadness (0.84), fear (0.81), anger (0.79), disgust (0.76), surprise (0.73), trust (0.70), anticipation (0.71), joy (0.82). The higher accuracy for high-intensity negative emotions reflects their greater linguistic distinctiveness and better representation in training datasets. The temporal analysis ablation study confirmed

a 12% sensitivity improvement on the longitudinal Reddit dataset subset (users with 5+ posts). Mean end-to-end latency: 3.2 seconds (SD=0.8s) under simulated production load. LLM API call accounts for 65% of total latency — the primary optimisation target for production deployment. Psychology graduate evaluators rated explanation clarity 4.2/5 and recommendation actionability 3.9/5.

5.5 ABLATION STUDY: CONTRIBUTION OF EACH PIPELINE COMPONENT

To quantify the individual contribution of each pipeline component to overall detection performance, an ablation study was conducted in which each component was systematically removed from the full pipeline while all others remained active. The following summarises the key findings:

Configuration	Sensitivity	Specificity	AUC-ROC	Notes
Full Pipeline (all components)	0.84	0.79	0.87	Baseline full-system performance.
Without Temporal Analysis	0.75	0.79	0.81	Most significant drop — confirms temporal analysis value.
Without Linguistic Markers	0.79	0.78	0.83	Moderate drop — markers add meaningful signal.
Without Hopelessness Factor	0.81	0.79	0.84	Small drop — hopelessness partially captured by emotion.
Without Social Factor	0.83	0.79	0.86	Minimal drop — social overlap with other features.
LLM only (no linguistic, no temporal)	0.72	0.80	0.80	Shows LLM alone is insufficient.

The ablation results clearly demonstrate that the temporal analysis component provides the largest individual contribution to detection performance — removing it causes a 6-point AUC-ROC drop from 0.87 to 0.81. The linguistic marker module contributes the next largest gain (0.04 AUC-ROC), while the hopelessness and social components add incremental improvements. Importantly, the LLM-only baseline (without linguistic markers or temporal analysis) achieves only AUC-ROC 0.80 — demonstrating that the full multi-component pipeline architecture is essential to the system's clinical effectiveness. These ablation findings validate the architectural decision to invest in all five pipeline components rather than relying on LLM inference alone.

5.6 USER EXPERIENCE AND CLINICAL COMMUNICABILITY EVALUATION

User experience and clinical communicability were assessed through structured evaluation sessions with two groups: (1) six graduate students in clinical psychology who evaluated assessment quality and recommendation appropriateness, and (2) eight non-specialist users who evaluated interface usability and

explanation accessibility. Key findings:

- **Explanation Clarity (Clinical Raters):** Mean 4.2/5 — raters appreciated the factor-ranked structure and natural language descriptions; noted that specific linguistic marker evidence could be more prominently displayed.
- **Recommendation Appropriateness (Clinical Raters):** Mean 3.9/5 — tiering logic was approved; Crisis tier resources were rated highly appropriate; some raters suggested adding professional telehealth platform links at High level.
- **Task Completion Rate (Non-Specialist Users):** 100% — all users successfully submitted text, received results, and located crisis resources without guidance.
- **Mean Time-on-Task:** 48 seconds from submission to dashboard review — rated as highly acceptable.
- **UI Satisfaction (Non-Specialist Users, 1-5 Likert):** Mean 4.4/5 — users appreciated the colour-coded risk categories and the emotion profile visualisation.
- **Perceived Trustworthiness (Non-Specialist Users):** Mean 3.8/5 — users found explainability reassuring but expressed desire for clearer disclaimers about system limitations.

CHAPTER 6

CHALLENGES FACED:

The development of the Emotion-Aware Mental Health Risk Detection system presented numerous significant technical and operational challenges. Each challenge required iterative problem-solving and deepened the team's understanding of building AI systems for high-stakes healthcare domains.

6.1 LLM PROMPT ENGINEERING CHALLENGES

Designing prompts that reliably elicit structured, clinically meaningful emotional analysis from GPT-4 proved among the most technically demanding aspects of the project. Initial prompt designs produced inconsistent output formats, varying levels of specificity, and occasional category confusion between similar emotions (sadness vs. fear). Achieving reliable JSON schema compliance required developing multi-attempt retry logic with progressively constrained prompt variations, rigorous output validation pipelines, and structured output mode configurations. The chain-of-thought prompting approach ultimately adopted required over 40 iterations of empirical testing across diverse test inputs spanning different emotional registers, cultural contexts, and text lengths.

6.2 EMOTION CLASSIFICATION ACCURACY CHALLENGES

Achieving uniformly high accuracy across all eight emotion categories proved inherently challenging. Subtle cognitive-affective emotions (anticipation, trust) are significantly underrepresented in training corpora compared to high-intensity emotions (sadness, anger), producing a clear performance gradient in per-category F1 scores. Mixed emotional states — co-occurring joy and sadness in bittersweetness, or fear and anticipation in anxiety — required careful multi-label probability calibration to prevent binary classification artifacts. Addressing these accuracy gaps required targeted few-shot examples in prompts specifically calibrated to underperforming emotion categories.

6.3 LINGUISTIC MARKER DETECTION CHALLENGES

Building a comprehensive, culturally sensitive, and clinically validated lexicon for absolutist terms, cognitive distortion vocabulary, and hedging language markers required extensive review of clinical psychology and computational linguistics literature. Balancing precision (avoiding false positives from benign emphatic language such as 'I absolutely love this') against recall (capturing genuine cognitive distortion patterns) required careful threshold calibration through empirical testing across diverse text samples. Context-dependent disambiguation of whether 'never' reflects clinical absolutism or casual hyperbole remains a fundamental challenge that semantic similarity approaches only partially address.

6.4 TEMPORAL ANALYSIS IMPLEMENTATION CHALLENGES

Designing MongoDB schema for efficient longitudinal per-user emotional records, computing meaningful personal baselines for new users with limited history using population-level priors, distinguishing genuine deterioration from normal day-to-day emotional fluctuation through robust statistical thresholds, and handling data gaps from irregular usage patterns without producing misleading trajectory estimates — all proved significant implementation challenges requiring iterative design refinement and rigorous testing with synthetic longitudinal data.

6.5 MULTI-FACTOR RISK SCORING CHALLENGES

Determining appropriate component weights without access to large-scale labelled clinical datasets with ground-truth severity ratings required calibration against existing benchmark datasets. Ensuring equitable performance across demographic groups — avoiding systematic over- or under-scoring for different cultural communication styles — required explicit bias analysis and weight adjustment. The T_{trend} adjustment magnitude required careful calibration to contribute meaningfully without overwhelming the base score for users with modest histories.

6.6 EXPLAINABILITY GENERATION CHALLENGES

Generating explanations simultaneously technically accurate, accessible and non-alarmist for general users unfamiliar with clinical terminology, and sufficiently detailed for clinical professional review presented a complex multi-objective communication challenge. Achieving this balance required iterative refinement with feedback from both psychology graduate students and lay user testers across multiple cycles of design and evaluation.

6.7 ETHICAL AND PRIVACY CHALLENGES

Implementing meaningful informed consent workflows that users genuinely understand rather than merely click through, ensuring the system does not inadvertently stigmatise users through false positive crisis alerts, designing crisis response pathways that are helpful without being prescriptive or alarming, and ensuring field-level encryption for mental health data throughout the full data lifecycle all required careful architectural and UX design decisions informed by mental health ethics frameworks.

6.8 TESTING AND DEBUGGING CHALLENGES

Testing a system with probabilistic LLM outputs required novel approaches beyond standard deterministic unit tests. Establishing ground truth labels required reliance on benchmark dataset annotations with inherent

label quality limitations. Debugging inter-module interactions required detailed component-level logging and systematic ablation studies isolating each pipeline stage's contribution to final assessment accuracy.

6.9 OVERALL DEVELOPMENT EXPERIENCE

Despite its challenges, the development of this system was a profoundly rewarding experience providing deep practical exposure to LLM prompt engineering, clinical NLP pipeline architecture, AI ethics frameworks, MongoDB schema design, FastAPI backend development, and React.js dashboard implementation — preparing the team for professional engineering careers in AI and healthcare technology.

CHAPTER 7

CONCLUSION:

The Emotion-Aware LLM for Early Mental Health Risk Detection project successfully presents a comprehensive, technically rigorous, and ethically grounded solution to one of the most critical unmet public health needs of our time. In an era where mental health conditions affect one in four people globally yet the majority go undetected until crisis points, AI-powered early detection systems that are sensitive, interpretable, and responsibly designed can play a transformative role in expanding the mental health safety net.

The system implements a sophisticated five-stage NLP pipeline: text preprocessing, LLM-based fine-grained emotion extraction across eight Plutchik categories, linguistic marker detection targeting five high-risk feature categories, temporal trajectory analysis maintaining longitudinal records per user, and weighted multi-factor risk scoring producing a transparent 0–100 assessment with confidence estimates. The React.js frontend delivers an intuitive dashboard; the FastAPI backend efficiently orchestrates the multi-stage pipeline; and MongoDB provides reliable longitudinal data persistence.

The temporal analysis integration proved to be among the project's most significant achievements, improving detection of gradually deteriorating mental states by 12% over single-point assessment — a clinically meaningful improvement given that serious mental health crises frequently develop gradually and are missed by systems lacking longitudinal perspective. The system achieves AUC-ROC 0.87, outperforming keyword-based (0.78) and traditional ML (0.82) baselines. Psychology graduate evaluators rated explanation clarity 4.2/5 and recommendation actionability 3.9/5, confirming clinical communicability of the system outputs.

The tiered recommendation engine — delivering proportionate support from wellness resources at Low risk to immediate crisis hotlines at Crisis level — ensures the system functions as a genuine mental health safety net rather than a diagnostic classifier alone. The ethical design incorporating informed consent, data encryption, access controls, and bias mitigation strategies establishes a responsible foundation for potential future clinical deployment.

7.1 LIMITATIONS AND ETHICAL CONSIDERATIONS

Despite the promising results, several important limitations must be transparently acknowledged:

Technical Limitations

- Assessment accuracy depends critically on input text quality and length; messages shorter than 50 words produce less reliable analyses.
- Performance may vary across languages, cultural contexts, and demographics not represented in English-language training datasets.
- LLM inference requires substantial computational resources and reliable internet connectivity.
- Temporal analysis requires a minimum of 3–5 historical assessments before meaningful trajectory analysis is possible.
- The system has not undergone formal clinical validation with licensed mental health professionals.

Ethical Considerations

- **Screening vs. Diagnostic Tool:** The system is explicitly designed as a supplementary screening tool to complement — not replace — professional clinical assessment.
- **False Positive/Negative Harms:** False positives may cause unnecessary anxiety; false negatives may miss genuinely at-risk individuals. Both error types are ethically significant and communicated transparently to users.
- **Informed Consent:** Users must provide meaningful informed consent explaining data collection, processing, retention, and the limitations of automated assessment before any text is analysed.
- **Data Privacy:** The highly sensitive nature of mental health data demands highest standards of security, encryption, and data minimisation throughout the data lifecycle.
- **Algorithmic Bias:** Systematic training data biases can produce inequitable performance across demographic groups; ongoing bias monitoring is essential for responsible deployment.
- **Human Oversight:** All consequential mental health care decisions must involve qualified human clinical judgment — the system must never remove human oversight from high-stakes intervention decisions.

7.2 PROJECT REFLECTIONS AND LESSONS LEARNED

The development of this system provided the project team with invaluable practical experience at the intersection of AI/ML engineering, clinical psychology, healthcare ethics, and full-stack software development. Several key lessons emerged that will inform future work in this domain.

First, the technical and ethical dimensions of healthcare AI are inseparable. Every design decision — from risk threshold calibration to explanation phrasing to crisis response UI design — carries ethical weight that cannot be addressed retrospectively. Embedding ethics-by-design from the outset of architecture planning, rather than treating it as a compliance checkbox, was both more effective and more intellectually satisfying.

Second, the temporal analysis component — which delivered the single largest accuracy improvement — would have been missed entirely had the project scope been limited to single-point text analysis as most prior work assumes. This reinforces a broader principle: in clinical domains, the history is often as diagnostically important as the current state, and AI systems designed for healthcare contexts should be architected with longitudinal data modelling as a first-class design requirement rather than an afterthought.

Third, explainability is not merely a technical feature but a fundamental requirement for clinical trust and adoption. The 4.2/5 clarity rating from psychology graduate evaluators demonstrates that LLM-generated natural language explanations can achieve clinically communicable quality — but this required substantial iterative prompt engineering and evaluation to achieve. Systems deploying AI in clinical contexts should budget significant development time and user testing effort for explainability design.

CHAPTER 8

FUTURE SCOPE:

While the current implementation demonstrates strong feasibility and effectiveness on benchmark evaluation, it represents the foundation of a much larger and more impactful system. The following sections describe planned and potential enhancements:

8.1 CLINICAL VALIDATION STUDIES

The most critical near-term enhancement is conducting rigorous clinical validation studies in collaboration with licensed psychiatrists, clinical psychologists, and mental health organisations. Real-world deployment with clinician oversight will enable calibration of risk thresholds against clinical ground truth severity ratings, validation of linguistic markers across diverse patient populations, refinement of recommendation logic based on professional clinical judgment, and generation of the peer-reviewed evidence base necessary for responsible broader deployment.

8.2 PERSONALIZED BASELINE MODELS

Future versions will develop fully individualised emotional baseline models per user, progressively trained on their growing longitudinal assessment history using Bayesian updating. Rather than comparing all users to population-level norms, personalised baselines will detect meaningful departures from each individual's characteristic emotional expression style — substantially reducing false positive rates.

8.3 CROSS-LINGUAL AND CROSS-CULTURAL SUPPORT

Extending support to major world languages through multilingual transformer models including mBERT, XLM-RoBERTa, and IndicBERT for South Asian language coverage. Critical to this expansion is adapting linguistic marker lexicons and risk thresholds to culturally specific emotional expression norms, idioms, and help-seeking conventions — requiring collaboration with mental health researchers from target language communities.

8.4 MULTIMODAL INPUT SUPPORT

Future versions will extend beyond text to support speech recordings (prosodic and acoustic feature extraction), facial expression video (micro-expression analysis), and wearable physiological data (HRV, galvanic skin response). Multimodal fusion will produce substantially more robust assessments — particularly for users who express themselves more readily through voice or in low-connectivity environments.

8.5 MOBILE APPLICATION DEVELOPMENT

A dedicated native mobile application using React Native or Flutter will improve accessibility and enable passive background emotional journaling prompts, push notification reminders for check-ins, offline assessment capability, location-aware context integration, and deeper integration with device health sensors.

8.6 REAL-TIME CRISIS INTERVENTION INTEGRATION

Implementing consent-based escalation pathways that — when crisis-level risk is detected with high confidence — can optionally alert designated trusted contacts or mental health professionals in near real-time. This requires careful privacy-preserving consent frameworks, legal compliance across jurisdictions, and integration with crisis intervention service APIs.

8.7 EHR INTEGRATION

Integration with Electronic Health Record systems via HL7 FHIR will enable longitudinal mental health screening data to be incorporated into patients' holistic medical records accessible to treating clinicians, enabling comprehensive patient-centred care coordination across primary care and specialist settings.

8.8 FEDERATED LEARNING FOR PRIVACY PRESERVATION

Federated learning architectures will enable continuous improvement of risk scoring models from distributed user interaction data without centralising sensitive mental health information — a privacy-preserving training paradigm essential for building user trust in mental health applications at scale.

8.9 EXPANSION OF CONDITION COVERAGE

Future versions will expand coverage from depression and anxiety to PTSD, bipolar disorder, eating disorders, substance use disorders, and OCD — each requiring specialised linguistic marker lexicons, condition-specific LLM prompts, and dedicated clinical validation partnerships.

8.10 CLOUD-BASED SCALABILITY

Migration to production-grade cloud infrastructure (AWS, Azure, or GCP) with elastic compute, GPU-accelerated LLM inference, and strategic caching will reduce mean response time below 2 seconds and support large-scale concurrent user deployment at population scale.

8.11 OVERALL FUTURE VISION

The long-term vision is transformation into a comprehensively validated, globally accessible, multimodal, and personalised mental health screening ecosystem that measurably reduces the gap between mental health need and timely professional intervention at population scale — with genuine potential to become a standard component of digital mental health infrastructure worldwide.

8.12 RESEARCH ROADMAP AND MILESTONES

To translate the current prototype into a clinically validated, production-ready system, the following phased research and development roadmap is proposed:

Phase 1: Clinical Validation (Year 1)

- Partner with 2–3 psychiatric institutions for supervised real-world deployment.
- Collect clinician-labelled assessment data for threshold calibration.
- Conduct IRB-approved user study with minimum 200 participants.
- Publish clinical validation findings in a peer-reviewed mental health journal.
- Incorporate clinical feedback to refine recommendation logic and risk tier thresholds.

Phase 2: Platform Expansion (Year 2)

- Develop and publish React Native mobile application for iOS and Android.
- Add Hindi and 5 additional regional Indian language support via IndicBERT.
- Integrate with iCall and Vandrevalla Foundation APIs for seamless crisis escalation.
- Implement federated learning framework for privacy-preserving model improvement.
- Pilot EHR integration with 1–2 partner healthcare institutions via HL7 FHIR.

Phase 3: Scale and Impact (Year 3+)

- Achieve 10,000+ registered users with longitudinal assessment data.
- Expand to multimodal input: speech and facial expression analysis modules.
- Extend condition coverage to PTSD, bipolar disorder, and substance use disorders.
- Pursue regulatory consultation for clinical decision support tool classification.
- Publish comprehensive impact study measuring real-world mental health outcomes.

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APPENDICES

APPENDIX A: KEY SOURCE CODE SNIPPETS

The following table summarises the key source files, their roles, and core logic in the Emotion-Aware Mental Health Risk Detection system.

Table A.1: Key Source Code Snippets of the Project

File / Module	Role in System	Key Logic / Functions
backend/auth/routes.py	User Auth	POST /register (bcrypt+JWT), POST /login, GET /me, PUT /profile
backend/pipeline/preprocess.py	Text Preprocessing	Unicode normalize, URL remove, expand contractions, emoticon map, SpaCy tokenise
backend/pipeline/emotion_extractor.py	LLM Emotion Extraction	Chain-of-thought GPT-4 prompt, JSON schema validation, retry logic, VADER fallback
backend/pipeline/linguistic_analyzer.py	Linguistic Markers	PronounDensity, AbsolutistScore, TemporalFocusRatio, SocialRefFreq, HedgingDensity
backend/pipeline/risk_scorer.py	Risk Scoring Engine	$RiskScore = \sum(w_i \cdot F_i) + T_trend$; normalise 0–100; category; confidence
backend/pipeline/temporal_analyzer.py	Temporal Analysis	LinReg slope on history; volatility; baseline deviation; T_trend adjustment
backend/pipeline/report_generator.py	Explainable Reports	Factor ranking; LLM-NLG explanations; temporal narrative; JSON report
backend/pipeline/recommender.py	Recommendations	Risk-level threshold lookup; tier selection; crisis hotline injection
frontend/src/pages/Dashboard.jsx	User Dashboard UI	Risk gauge, emotion bars, factor chart, recommendations panel, temporal sparkline
frontend/src/hooks/useAssessment.js	Assessment Hook	Submit text to API; handle loading/error state; refresh dashboard

APPENDIX B: SYSTEM REQUIREMENTS SPECIFICATION

B.1 Hardware Requirements

Table B.1: Hardware Requirements for the System

Component	Minimum Requirement	Recommended
Processor	Intel Core i5 / AMD Ryzen 5	Intel Core i7+ / NVIDIA GPU
RAM	8 GB	16 GB or above
Storage	256 GB SSD	512 GB SSD or above
Internet	Broadband 4G	High-speed / 5G Fibre
GPU (local LLM)	4 GB VRAM	12 GB+ VRAM (NVIDIA RTX series)

B.2 Software Requirements

Table B.2: Software Requirements for the System

Software / Tool	Version	Purpose
Python	3.10+	Backend Runtime & NLP Pipeline
FastAPI	0.100+	REST API Framework
PyTorch	2.x	Deep Learning Framework
HuggingFace Transformers	4.35+	Pre-trained LLM Models
OpenAI Python SDK	1.x	GPT-4 API Integration
React.js	18.2	Frontend Framework
MongoDB (Atlas)	6.x	NoSQL Database
VS Code	1.85	Integrated Development Environment
Postman	v10.x	API Testing and Debugging
Git / GitHub	2.x	Version Control and Collaboration

APPENDIX C: REST API ENDPOINT REFERENCE*Table C.1: REST API Endpoint Reference for the Backend Server*

Method	Endpoint	Auth ?	Description
POST	/api/auth/register	No	Register new user account
POST	/api/auth/login	No	Authenticate user; return JWT bearer token
GET	/api/user/profile	JWT	Retrieve logged-in user profile
PUT	/api/user/profile	JWT	Update user profile details
POST	/api/assess/submit	JWT	Submit text for full mental health risk assessment
GET	/api/assess/history	JWT	Retrieve user's past assessment records
GET	/api/assess/{id}/report	JWT	Get detailed explainable report for assessment
GET	/api/temporal/trend	JWT	Get user's emotional trajectory analytics
POST	/api/feedback/submit	JWT	Submit feedback rating for recommendations
GET	/api/resources/{level}	JWT	Get mental health resources for given risk level

PUBLICATIONS

International Journal of Computer Techniques (IJCT)

ISSN: 2394-2231 | www.ijctjournal.org | Peer-Reviewed Open Access



Paper Acceptance Notification

Dear Authors,

We are pleased to inform you that your paper titled:

"Emotion-Aware LLM for Early Mental Health Risk Detection"

Review Summary

Round 1: Plagiarism & Similarity Check	✓ Passed
Round 2: Initial Editorial Review	✓ Passed
Round 3: External Reviewer Assessment	✓ Passed
Round 4: Author Revisions & Final Evaluation	✓ Passed
Round 5: Formatting & Template Compliance	✓ Passed
Round 6: Technical Quality & Language Review	✓ Passed
Round 7: Indexing & DOI Verification	✓ Approved

Manuscript cleared all review stages — publication quality confirmed.

P a g e | **41**

Dupli Checker

International Journal of Computer Techniques (IJCT) | ISSN: 2394-2231

Plagiarism Report | Date: 06-05-2026 | Report No: IJCT-2026-0512

✓ Approved

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Plagiarism

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Speak Time:68 min

Paragraphs:785

Read Time:51 min

Sentences:841

Characters:7046

Words:1,000

Document Statistics

APPROVED — Content cleared all review stages. Similarity within acceptable limit.

CURRICULUM VITAE

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Career Objective

Motivated B.Tech Computer Science Engineering student with a strong foundation in Natural Language Processing, Machine Learning, and Full-Stack Web Development. Proficient in Python, Java, and JavaScript with hands-on experience designing and building AI-powered applications. Passionate about applying AI to healthcare and social-good domains to create meaningful, measurable real-world impact. Seeking a challenging role in AI/ML engineering or research upon graduation.

Education

Noida Institute Of Engineering and Technology, Greater Noida	2022–Present
B.Tech Computer Science and Engineering CGPA: 8.56	
Delhi Public School, New Delhi	2020–2021
Intermediate (Class XII) Percentage: 91.2% Stream: Science (PCM + CS)	
Delhi Public School, New Delhi	2018–2019
High School (Class X) Percentage: 94.8% CBSE Board	

Technical Skills

- Programming Languages: Python, Java, JavaScript (ES6+), SQL, Bash
- ML / AI Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, Scikit-Learn, NLTK, SpaCy, VADER
- LLM & NLP: GPT-4 API, LangChain, Prompt Engineering, Chain-of-Thought, BERT, RoBERTa
- Frontend Development: React.js, HTML5, CSS3, Tailwind CSS, Axios
- Backend Development: FastAPI, Node.js, Express.js, REST API Design
- Databases: MongoDB, MySQL, Redis
- DevOps / Tools: Git, GitHub, Docker (basics), VS Code, Postman, Jupyter Notebook

Certifications & Online Courses

- Deep Learning Specialisation (5 courses) — Coursera / deeplearning.ai
- Natural Language Processing with Attention Models — Coursera / deeplearning.ai
- Python for Data Science and AI — IBM / Coursera
- Machine Learning with Python — IBM / Coursera
- Introduction to TensorFlow for AI, ML, and DL — Coursera

Aman Bajpai — continued

Projects & Contributions

Emotion-Aware LLM Mental Health Risk Detection (Major Project)

Full-stack AI mental health screening system published in IJCT. Led the entire system design, LLM prompt engineering, risk scoring engine, and integration.

- Designed and iterated the chain-of-thought GPT-4 prompting strategy that achieved reliable 8-category Plutchik emotion extraction.
- Implemented the weighted multi-factor risk scoring engine: $\text{RiskScore} = \sum(w \cdot F) + T_{\text{trend}}$ achieving AUC-ROC 0.87.
- Led API integration, end-to-end testing, and deployment of the FastAPI backend.
- Coordinated full team workflow, milestone tracking, and final report authorship.

BERT Sentiment Analysis Dashboard

End-to-end BERT fine-tuned sentiment classification pipeline on 50K Twitter samples with real-time React.js visualisation.

- Fine-tuned BERT-base-uncased on Sentiment140 dataset achieving 88% test accuracy.
- Built FastAPI serving layer with <200ms response time for real-time dashboard updates.
- Visualised live emotion trends with recharts-based interactive React.js dashboard.

Co-Authorship / Publications

- Co-authored: "**Emotion-Aware LLM for Early Mental Health Risk Detection**" — published in International Journal of Computer Techniques (IJCT), ISSN: 2394-2231, Vol. 13, Issue 2, May 2026.

Co-Curricular Activities & Achievements

- Member, NIET AI & Coding Club — organised 3 workshops on Python ML tools.
- Participated in HackerRank 30-day Python challenge — completed all 30 problems.
- Volunteer, NIET Technova Tech Fest 2024 — managed AI/ML track judging logistics.
- Completed NPTEL online course: Introduction to Machine Learning (IIT Madras) — Elite certificate.
- Active contributor to team GitHub repository with 200+ commits across the major project.

Professional Summary

Aman brings a comprehensive blend of theoretical AI/ML knowledge and strong practical engineering skills developed through the Major Project and independent coursework. His leadership of the LLM engineering

and system integration aspects of the mental health detection project demonstrates readiness for professional AI engineering roles in healthcare, NLP, or product AI teams.

Declaration

I hereby declare that all the information provided in this curriculum vitae is true and correct to the best of my knowledge and belief.

Signature: _____ Date: _____

Esha Kumari Singh Roll No: 2201330100103

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Career Objective

Aspiring AI/ML Engineer with strong foundations in Machine Learning, NLP, and data-driven software development. Proficient in Python, TensorFlow, and HuggingFace Transformers with hands-on experience building and evaluating NLP pipelines and deep learning models. Passionate about applying AI to healthcare and clinical decision support domains. Seeking an AI engineering or data science role that allows application of clinical NLP expertise to drive social impact.

Education

Noida Institute Of Engineering and Technology, Greater Noida	2022–Present
<i>B.Tech Computer Science and Engineering CGPA: 8.43</i>	
Kendriya Vidyalaya, New Delhi	2020–2021
<i>Intermediate (Class XII) Percentage: 89.6% Stream: Science (PCM + Biology)</i>	
Kendriya Vidyalaya, New Delhi	2018–2019
<i>High School (Class X) Percentage: 92.4% CBSE Board</i>	

Technical Skills

Programming Languages: Python, R, SQL, JavaScript (basic)

ML / AI Frameworks: TensorFlow, PyTorch (basic), Scikit-Learn, HuggingFace Transformers, Keras

NLP Tools: NLTK, SpaCy, VADER, LIWC-style lexicon construction, BERT, RoBERTa

Data Analysis: Pandas, NumPy, Matplotlib, Seaborn, SciPy, Jupyter Notebook

Backend & APIs: FastAPI, REST API integration, JSON schema design

Databases: MongoDB, PostgreSQL, SQLite

Tools & Platforms: Git, GitHub, Google Colab, VS Code, Kaggle

Certifications & Online Courses

- Machine Learning Specialisation (3 courses) — Coursera / Stanford (Andrew Ng)
- AI for Medical Diagnosis — Coursera / deeplearning.ai

- SQL Intermediate Certificate — HackerRank
- Data Analysis with Python — IBM / Coursera
- Introduction to Deep Learning & Neural Networks — Coursera

Esha Kumari Singh — continued

Projects & Contributions

Emotion-Aware LLM Mental Health Risk Detection (Major Project)

Designed and implemented the linguistic marker detection module and complete temporal pattern analysis system for the NIET B.Tech Major Project.

- Built the 5-category linguistic marker extractor: pronoun density, absolutist language, temporal focus ratio, social reference, hedging density.
- Designed MongoDB schema for efficient longitudinal per-user assessment storage with compound indexing.
- Implemented temporal trajectory analysis: linear regression slope, volatility index, baseline deviation, T_trend computation.
- Led benchmark evaluation on CLPsych dataset and co-authored the IJCT research paper.

Medical Image Classification (CNN)

Chest X-ray pulmonary disease detection system using CNN with data augmentation to handle class imbalance.

- Implemented ResNet-50 fine-tuning on NIH Chest X-ray dataset achieving 89% macro F1.
- Applied SMOTE oversampling and augmentation (flips, rotations, contrast jitter) for class balance.
- Evaluated with AUC-ROC curves and confusion matrices across 14 pathology classes.

Co-Authorship / Publications

- Co-authored: "**Emotion-Aware LLM for Early Mental Health Risk Detection**" — published in International Journal of Computer Techniques (IJCT), ISSN: 2394-2231, Vol. 13, Issue 2, May 2026.

Co-Curricular Activities & Achievements

- Member, NIET Women in Tech Cell — participated in panel discussion on AI in healthcare.
- Completed Kaggle Titanic and Housing Price competitions — ranked top 30%.
- NPTEL course: Joy of Computing using Python (IIT Ropar) — Elite + Gold certificate.
- Volunteer, NIET Blood Donation Camp coordination — social responsibility initiative.
- Regular participant in AI/ML reading group — presented paper review on MentalBERT architecture.

Professional Summary

Esha brings deep NLP and data science skills, particularly in linguistic feature engineering and temporal data analysis. Her implementation of the linguistic marker detection and temporal analysis modules — which delivered the 12% sensitivity improvement — demonstrates strong applied ML engineering capability and clinical NLP expertise well-suited to AI in healthcare or data science roles.

Declaration

I hereby declare that all the information provided in this curriculum vitae is true and correct to the best of my knowledge and belief.

Signature: _____ Date: _____

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Career Objective

Results-driven Full-Stack Engineer and B.Tech CS student with expertise in React.js, Node.js, FastAPI, and MongoDB. Strong problem-solving skills and experience designing scalable RESTful APIs and responsive user interfaces. Adept at collaborating in agile team environments. Seeking a full-stack development or front-end engineering role where I can contribute to building impactful, user-centred products.

Education

Noida Institute Of Engineering and Technology, Greater Noida	2022–Present
B.Tech Computer Science and Engineering CGPA: 8.35	
Army Public School, New Delhi	2020–2021
Intermediate (Class XII) Percentage: 90.4% Stream: Science (PCM + CS)	
Army Public School, New Delhi	2018–2019
High School (Class X) Percentage: 93.1% CBSE Board	

Technical Skills

- Programming Languages:** JavaScript (ES6+), Python, Java, TypeScript (basic)
- Frontend Development:** React.js, HTML5, CSS3, Tailwind CSS, Redux, Axios, Vite
- Backend Development:** Node.js, Express.js, FastAPI, REST API design, JWT Auth, bcrypt
- Database & Storage:** MongoDB (Mongoose), MySQL, Redis (basic), MongoDB Atlas
- UI/UX Design:** Figma (prototyping), responsive design, accessibility (WCAG basics)
- DevOps / Tooling:** Git, GitHub, Docker (basics), Postman, VS Code, Vercel, Render

Certifications & Online Courses

- React – The Complete Guide 2024 — Udemy (Maximilian Schwarzmüller — 40h course)
- Full Stack Web Development with MERN Stack — Udemy
- Object-Oriented Programming in Java — Coursera / UC San Diego
- Node.js, Express, MongoDB & More: The Complete Bootcamp — Udemy

- JavaScript Algorithms and Data Structures — freeCodeCamp (300h)

Ayush Raj — continued

Projects & Contributions

Emotion-Aware LLM Mental Health Risk Detection (Major Project)

Designed and developed the complete React.js frontend dashboard and all backend REST API integrations for the NIET Major Project.

- Built the full React.js assessment dashboard: risk score gauge, emotion profile bar chart, factor contribution panel, temporal sparkline, recommendations panel.
- Implemented all FastAPI REST endpoints (/auth, /assess, /temporal, /feedback, /resources) with JWT middleware.
- Designed and implemented MongoDB schema with compound indexes for efficient longitudinal user data queries.
- Integrated OpenAI GPT-4 API with error handling, retry logic, and streaming response parsing.

E-Commerce Platform (MERN Stack Full-Stack)

Full-featured e-commerce web application with complete shopping and order management workflow.

- React.js frontend with product catalogue, cart, wishlist, and responsive mobile-first design.
- Node.js/Express.js REST API backend with JWT authentication, Mongoose ODM, and Razorpay payment gateway integration.
- Admin dashboard for product management, order tracking, and user management with role-based access control.
- Deployed on Vercel (frontend) + Render (backend) + MongoDB Atlas (database) — fully cloud-hosted.

Co-Authorship / Publications

- Co-authored: "**Emotion-Aware LLM for Early Mental Health Risk Detection**" — published in International Journal of Computer Techniques (IJCT), ISSN: 2394-2231, Vol. 13, Issue 2, May 2026.

Co-Curricular Activities & Achievements

- Member, NIET Web Dev & Open Source Club — contributed UI to 2 club open-source projects.
- Completed 30-day JavaScript challenge on freeCodeCamp — 300+ hours of coursework.
- Participated in NIET internal hackathon — built a task management app in 24 hours (top 5 finish).
- Volunteer web developer — built informational website for local NGO using React.js and Tailwind CSS.

- Active on GitHub — 150+ public commits across personal and team projects.

Professional Summary

Ayush is a proficient full-stack engineer who designed and implemented the entire user-facing layer of the mental health detection project — from the React.js interactive dashboard to the FastAPI backend and MongoDB schema. His end-to-end ownership of the frontend/backend integration demonstrates strong production-ready engineering skills ideal for full-stack or front-end engineering roles.

Declaration

I hereby declare that all the information provided in this curriculum vitae is true and correct to the best of my knowledge and belief.

Signature: _____ Date: _____